Department of Biochemistry

UG PROGRAMME

Choice Based Credit System – Learning Outcomes Based Curriculum Framework (CBCS-LOCF)

(Applicable to the candidate admitted from the academic year 2025 – 2026 onwards)

BOS - I

E. G. S. PILLAY ARTS & SCIENCE COLLEGE (AUTONOMOUS)
NAGAPATTINAM – 611 002. TAMIL NADU
(Accredited by NAAC with "A" Grade)
(MBA – Approved by AICTE, New Delhi)
(Affiliated to Bharathidasan University, Tiruchirappalli – 24)

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							Int.	Ext.	Total
I	I	Language Course–I (Tamil$/ Other Languages+#)	Tamil-I	6	3	3	25	75	100
	II	English Language Course – I(ELC)	English -I	6	3	3	25	75	100
	III	Core Course – I (CC)	Biomolecules	5	5	3	25	75	100
		Corse Practical – I (CP)	Biomolecules (P)	4	4	3	50	50	100
		First Allied Course – I (AC)	Chemistry – I	4	3	3	25	75	100
		First Allied Practical –I(AP)	Allied Chemistry –I (P)	3	-	-	-	-	-
	IV	Value Education	Value Education	2	2	3	25	75	100
			Total	**30**	**20**				600

Sem.	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
II	I	LanguageCourse–II(Tamil$/ Other Languages+#)	Tamil- II	6	3	3	25	75	100
	II	English Language Course-II (ELC)	English -II	6	3	3	25	75	100
	III	Core Course- II (CC)	Cell Biology	5	5	3	25	75	100
		Core Practical – II (CP)	Cell Biology (P)	4	4	3	50	50	100
		First Allied Course II(AC)	Chemistry – II	4	3	3	25	75	100
		First Allied Practical – II (AP)	Allied Chemistry – II (P)	3	2	3	50	50	100
	IV	Environmental studies	Environmental Studies	2	2	3	25	75	100
			Total	**30**	**22**				**700**

K. UMA, M.Sc., M.Phil.,
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Dr. C. PRAHALATHAN, M.Sc., Ph.D.
Professor, Department of Biochemistry,
Bharathidasan University Tiruchirappalli – 620 024.

Sem	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
III	I	Language Course–III (Tamil$/ Other Languages+#)	Tamil-III	6	3	3	25	75	100
	II	English language Course-III (ELC)	English -III	6	3	3	25	75	100
	III	Core Course-III (CC)	Biochemical Techniques	5	5	3	25	75	100
		Core Practical – III (CP))	Biochemical Techniques (P)	4	4	3	50	50	100
		Second Allied Course-I (AC)	Microbiology - I	4	3	3	25	75	100
		Second Allied Practical –I (AP)	Allied Microbiology - I (P)	3	-	-	-	-	-
	IV	Non Major Elective –I (NME)	(A) Medicinal Diet (B) Health & Nutrition	2	2	3	25	75	100
		Activity Based Course (ABC)	Health and Wellness	-	1	-			100
			Total	**30**	**21**				**700**

Sem	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
IV	I	Language Course – IV (LC)) Tamil /Other Language	Tamil-IV	6	3	3	25	75	100
	II	English Course-IV (ELC)	English -IV	6	3	3	25	75	100
	III	Core Course-IV (CC)	Human Physiology	5	5	3	25	75	100
		Core Practical –IV (CC)	Human Physiology(P)	4	4	3	50	50	100
		Second Allied Course – II(AC)	Microbiology – II	4	3	3	25	75	100
		Second Allied Practical – II (AP)	Allied Microbiology - II (P)	3	2	3	50	50	100
	IV	Non – Major Elective Course- II (NME)	(A) First aid &Home Nursing (B) Life style disease	2	2	3	25	75	100
			Total	**30**	**22**				**700**

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Sem.	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
V	III	Core Course – V (CC)	Enzymes	6	5	3	25	75	100
		Core Course – VI (CC)	Molecular Biology	6	5	3	25	75	100
		Core Course – VII (CC)	Intermediary Metabolism	6	5	3	25	75	100
		Core Practical – V (CP)	Enzymes & Molecular Biology (P)	4	4	3	50	50	100
		Major Based Elective – I (MBE)	**(A)**Nutritional Biochemistry **(B)**Biotechnology	4	4	3	25	75	100
	III	Internship	Internship Report of the study= 80 marks Viva=20 marks	-	2	-	20	80	100
	IV	Skill Based Elective –I (SBE)	**(A)** Biomedical Instumentation **(B)** Basics of forensic science	2	2	3	25	75	100
	V	Soft Skill Development	Soft Skill Development	2	2	3	25	75	100
			Total	**30**	**29**				**800**

Sem.	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
							Int.	Ext.	Total
VI	III	Core Course –VIII (CC)	Immunology	6	6	3	25	75	100
		Core Course – IX (CC)	Clinical Biochemistry	6	5	3	25	75	100
		Core Practical– VI (CP)	Clinical Biochemistry &Immunology(P)	4	4	3	50	50	100
		Major Based Elective Course –II (MBE)	**(A)** Endocrinology **(B)** Bioinformatics	5	4	3	25	75	100
		Project	Project work Dissertation =80 marks Viva =20 marks	6	3	-	20	80	100
	IV	Skill Based Elective Course-II(SBE)	**(A)** Medical Lab Technology **(B)** Medical coding	2	2	3	25	75	100
	V	Gender Studies	Gender Studies	1	1	3	25	75	100
		Extension Activities	NCC,NSS,YRC,YCF	-	1	-	-	-	-
			Total	**30**	**26**				**700**
	GRAND TOTAL			**180**	**140**				**4200**

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NME: I Semester – III

A) Medicinal Diet

B) Health &Nutrition

NME: II Semester - IV

A) First aid and Home Nursing

B) Life style diseases

SBE: I – Semester - V

A) Biomedical Instrumentation

B) Basics of forensic science

SBE: II – Semester - VI

A) Medical Lab Technology

B) Medical coding

MBE: I-Semester-V

A) Nutritional Biochemistry

B) Biotechnology

MBE: II-Semester-VI

A) Endocrinology

B) Bioinformatics

SUMMARY OF CURRICULUM STRUCTURE OF UG PROGRAMME- BIOCHEMISTRY

Sl.No.	Part	Types of the Courses	No. of Courses	No.of Credits	Marks
1.	I	Languages Courses	04	12	400
2.	II	English Courses	04	12	400
3.	III	Core courses	09	46	900
4.		Core practicals	06	24	600
5.		Allied courses – I & II	04	12	400
6.		Allied Practical	02	04	200
7.		Major Based Elective Courses	02	08	200
8.		Project	01	03	100
9.		Internship	01	02	100
10.	IV	Non – Major Elective courses	02	04	200
11.		Skill Based Elective Courses	02	04	200
12.		Soft Skills Development	01	02	100
13.		Value education	01	02	100
14.		Environment studies	01	02	100
15.	V	Gender Studies	01	01	100
16.		Extension Activities	01	01	-
17.		Activity based course	01	01	100
Total			**43**	**140**	**4200**

Instructions:

$	For those who studied Tamil upto 10th +2 (Regular Stream)
+	Syllabus for other languages should be on par with Tamil at degree level
#	Those who studied Tamil upto 10th +2 nut opt for other languages in degree level under Part- I should study special Tamil in Part – IV
@	NCC Course is one of the Choices in Non- Major Elective Course. Only the NCC cadets are eligible to choose this course. However, NCC Course is not a Compulsory course for the NCC Cadets.
**	Extension Activities shall be instruction hours.

Programme Outcomes

The B.Sc. Biochemistry program describe accomplishments that graduates are expected to attain within five to seven years after graduation.

- An ability to apply fundamental knowledge related to sciences in an interdisciplinary manner for providing innovative solutions to need based problems for global impact.
- An ability to critically analyze scientific data, draw objective conclusions and apply this knowledge for human welfare. Students should be able to demonstrate expertise and ethical perspective on areas related to Biochemistry.
- An ability to gain domain knowledge and know-how for a successful career in academia, industry and research. Promoting lifelong learning to meet the ever evolving professional demands by developing ethical, inter personal and team skills.

Programme Specific Outcomes

After the successful completion of B.Sc. Biochemistry program, the students are expected to

- Broad based knowledge in biochemistry.
- Ability to understand the technical aspects of existing technologies that help in addressing the biological and medical challenges faced by humankind. Ability to contribute effectively in the development of the ethical practices, societal contributions, and leading to responsible and competent professionals.
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- Acquiring the ability of leadership skills to manage projects in multidisciplinary environments.
- To compete globally with confidence in all the sectors of life science.

BOS - II

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	III	Core Course – I (CC)	Biomolecules	5	5	3	25	75	100
		Corse Practical – I (CP)	Biomolecules (P)	4	4	3	50	50	100
		First Allied Course – I (AC)	Chemistry – I	4	3	3	25	75	100
		First Allied Practical –I(AP)	Allied Chemistry –I (P)	3	-	-	-	-	-
	IV	Value Education	Value Education	2	2	3	25	75	100
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Sem.	Part	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
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II	I	LanguageCourse–II(Tamil$/ Other Languages+#)	Tamil- II	6	3	3	25	75	100
	II	English Language Course-II (ELC)	English -II	6	3	3	25	75	100
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		Core Practical – II (CP)	Cell Biology (P)	4	4	3	50	50	100
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20/05/2026

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11.		Skill Based Elective Courses	02	04	200
12.		Soft Skills Development	01	02	100
13.		Value education	01	02	100
14.		Environment studies	01	02	100
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- Ability to contribute effectively in the development of the ethical practices, societal contributions, and leading to responsible and competent professionals
- Acquiring the ability of leadership skills to manage projects in multidisciplinary environments
- To compete globally with confidence in all the sectors of life science.

SEMESTER – I

FIRST YEAR	CORE COURSE - I	CREDIT : 5
SEMESTER - I	BIOMOLECULES	25BCCC1

COURSE OUTCOMES:

- This course would help the students learn the structure and function relationship of biomolecules *viz.* Carbohydrates, Proteins & Lipids
- Also their importance with regard to maintenance and perpetuation of the living systems.

COURSE SPECIFIC OUTCOMES

- Upon completion of this course, the student will be able to:
- Easily understand the basic concepts/functions of solutes, chemical bonding and organic compounds.
- Describe the classification of biomolecules.
- Describe the basic reaction types and mechanisms of biomolecules.
- Understand the structures and functions of biomolecules.
- Analyse and study the chemical and biochemical properties of biomolecules.

UNIT-I **Hours 15**

Carbohydrates-Classification and biological significance. Monosaccharides: Structure and structural elucidation of glucose. Disaccharides: Structure and properties of disaccharides (lactose and sucrose). Polysaccharides: Homopolysaccharides - Occurrence, structure and biological significance of starch and glycogen. Heteropolysaccharides - Structure and biological significance of - hyaluronic acid and heparin.

UNIT-II **Hours 15**

Amino acids -Classification based on composition of side chain and nutritional significance.General structure of amino acids. 3 - and 1- letter abbreviations. Physical properties of amino acids, isoelectric point and optical activity. Chemical reactions due to carboxyl group, amino group and side chains. Colour reactions of amino acids.

UNIT-III **Hours 15**

Proteins-Classification based on shape, composition, solubility and functions. Properties of proteins - Ampholytes, isoelectricpoint, salting in and salting out, denaturation and renaturation. Levels of organization of protein structure- Primary structure, formation of peptide bond. Secondary structure-α helix, β- pleated sheet. Tertiary structure – with reference to myoglobin. Quaternary structure with reference to haemoglobin.

UNIT-IV **Hours 15**

Lipids- Lipids: General classification, chemical nature and biological functions. Fatty acids: classification, nomenclature, structure and properties of fatty acids and triglycerides, Characterization of fats- iodine value, saponification value and acid number. Compound lipids-Structure and functions of phospholipids and glycolipids. Derived lipids-Structure and functions of cholesterol.

UNIT-V **Hours 15**

Nucleic acids-Structure of purine and pyrimidine bases, nucleosides and nucleotides and their biological importance. Types of DNA: A, B, C and Z DNA. Types of RNA: mRNA, tRNA and rRNA. Properties of DNA-Hypochromic and hyperchromic effect, melting temperature, viscosity. Denaturation and annealing.

TEXT BOOKS:

1. Fundamentals of Bichemistry, J.L. Jain, 2005.
2. Essential of Biochemistry, U.Satyanarayana and U. Chakrapani, 2002.
3. Fundamentals of Biochemistry, Agarwal O.P., Goel Publishing House, 2008, 11th edition.
4. Fundamentals of Biochemistry, AmbikaShanmugam, 8th edition, 2016.
5. Textbook of Biochemistry, West & Todd.4th edition, Macmillan, 1966.

REFERENCE BOOKS:

1. Principles of Biochemistry, Lehninger, Nelson Cox Macmillan worth Publishers, 2013, 7th edition.
2. Harper's Biochemistry 29th edition, McGraw Hill, 2012.
3. Chemistry of Biomolecules, S. P. Bhutani, 2010.
4. Introductory Experiments on Biomolecules and their Interactions, Robert K. 2015.

BOS CHAIRMAN

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FIRST YEAR	CORE PRACTICAL - I	CREDIT : 4
SEMESTER - I	BIOMOLECULES (P)	25BCCP1

COURSE OUTCOMES:

The main objectives of this course are to

- Identify the biomolecules carbohydrates and amino acids by qualitative test
- Determine the quality of Lipids by titrimetric methods
- Isolate nucleic acids from plant and animal source

COURSE SPECIFIC OUTCOMES;

- Qualitatively analyze the carbohydrates and report the type of carbohydrate based on specific tests.
- Qualitatively analyze amino acids and report the type of amino acids based on specific tests.
- Determine the Saponification, Iodine and acid number of edible oil.
- Isolate the nucleic acid from biological sources.

I Qualitative analysis of carbohydrates HOURS: 60

1) Carbohydrates
   a) Glucose b) Fructose c) Maltose d) Sucrose e) Lactose f) Starch
2) Amino acids
   a) Arginine b) Cysteine c) Histidine d) Proline e) Tryptophan f) Tyrosine

II Titrimetric methods

1) Determination of Saponification value of an edible oil
2) Determination of Iodine number of an edible oil
3) Determination of Acid number of an edible oil

III. Group Experiments

1) Isolation of DNA from plant/animal source. (onion root and goat liver)
2) Isolation of RNA from rich source (Yeast and goat liver).

TEXT BOOKS

1. David T Plummer, An Introduction to Practical Biochemistry, 3rd edition, Tata McGraw-Hill Edition
2. J. Jayaraman Laboratory Manual in Biochemistry New Age International (P) Limited Fifth edition 2015
3. S.Sadasivam,A.Manickam Biochemical Methods New age International Pvt Ltd publisher's third edition 2018

REFERENCE BOOKS

1. Rageeb, Kiran Patil, M. BakshiRahman, Sufiyan Ahmad Raees A Practical book on Biochemistry Everest publishing house1st Edition, 2019
2. Introductory practical Biochemistry – S.K. Sawhney, Randhir Singh, 2nd ed, 2005.
3.Biochemical Tests – Principles and Protocols. Anil Kumar, SarikaGarg and NehaGarg.VinodVasishtha Viva Books Pvt Ltd, 2012.
4. Harold Varley, Practical Clinical Biochemistry, CBS. 6 edition, 2006.
5. Keith Wilson and John Walker. Principles and Techniques of Practical Biochemistry, 4thedition, Cambridge University press, Britain.1995.

BOS CHAIRMAN

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SEMESTER – II

FIRST YEAR	CORE COURSE - II	CREDIT : 5
SEMESTER - II	CELL BIOLOGY	25BCCC2

COURSE OUTCOMES:

The main objectives of this course are to

- Provide basic understanding of architecture of cells and its organelles.
- Understand the organization of prokaryotic and eukaryotic genome.
- Educate on the structural organization of bio membrane and transport mechanism
- Impart knowledge on cell cycle, cell division and basics of cells
- Familiarize the concept of mechanism of cell-cell interactions.

COURSE SPECIFIC OUTCOMES :

- Explain the structure and functions of basic components of prokaryotic and eukaryotic cells, especially the organelles.
- Familiarize the cytoskeleton and chromatin.
- Illustrate the structure, composition and functions of cell membrane related to membrane transport
- Elaborate the phases of cell cycle and cell division-mitosis and meiosis and characteristics of cancer cells.
- Relate the structure and biological role of extra cellular matrix in cellular interactions

UNIT-I **Hours 15**

Architecture of cells- Structural organization of prokaryotic and eukaryotic cells microbial, plant and animal cells. Bio membranes-Structural organization of bilipid layer model and basic functions- transport across cell membranes- uniport, symport and antiport. Passive and active transport.

UNIT-II **Hours 15**

Structure and Functions of the ultrastructure of nucleus, mitochondria, RER, SER, golgi apparatus, lysosome, peroxisome and their functions.

UNIT-III **Hours 15**

Cytoskeleton- microfilament, microtubules and intermediary filament- structure, composition and functions. Organization of Genome -prokaryotic and eukaryotic genome. Organization of chromatin – histones, nucleosome concept, formation of chromatin structure. Special types of chromosomes – lamp brush chromosomes, polytene chromosomes.

UNIT-IV **Hours 15**

Extracellular matrix – Collagen, laminin, fibronectin and proteoglycans- structure and biological role. Structure and role of cadherin, selectins, integrins, Cell -cell interactions- Types-gap junctions, tight junctions and Desmosomes

UNIT-V **Hours 15**

Cell cycle-Definition and Phases of Cell cycle-Cell division-Mitosis and Meiosis and its significance, Cancer cells- definition, types and characteristics of cancer cells.

TEXT BOOKS

1. Arumugam. N, Cell biology. Saras publication (10ed, paperback), 2019
2. Devasena.T. Cell Biology. Oxford University Press India-ISBN: 9780198075516, 0198075510, 2012
3. Bruce Alberts and Dennis Bray. 2013, Essential Cell Biology. (4"ed). Garland Science.

REFERENCE BOOKS

1.S.C,R.Cell Biology. Newage Publishers -ISBN-10: 8122416888/ISBN-13: 978-8122416886, 2008
2.Cooper, G.A .The Cell: A Molecular Approach. Sinauer Associates, Inc -ISBN10: 0878931066 / ISBN 13: 9780878931064, 2013
3.E.M.F.,D.R, Cell and Molecular Biology. Lippincott Williams & Wilkins Philadelphia - ISBN: 0781734932 9780781734936, 2006
4. LodishH.A, Berk C.A, Kaiser M, Krieger M.P, Scott A, Bretscher H, Ploegh and Matsudaira. 2007. Molecular Cell Biology, 6th Edition, WH. Freeman Publishers, New York, USA.

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FIRST YEAR	CORE PRACTICAL - II	CREDIT : 4
SEMESTER - II	CELL BIOLOGY (P)	25BCCP2

COURSE OUTCOMES :

The main objectives of this course are to

- Learn the parts of microscope
- Investigation of cells
- At the cells under microscope.
- Image the cells using different stains
- Identify the cells, organelles and stages of cell division
- Identify the spotters
-

COURSE SPECIFIC OUTCOMES:

- Identify the tissues and Cells parts of microscope
- Preparation of Slides
- Identify the stages of mitosis & meiosis.
- Visualize nucleus and mitochondria by staining methods.
- Identify the spotters of cells, organelles and stages of cell division

I MICROSCOPY AND STAINING TECHNIQUES HOURS : 60

1. Parts of microscope
2. Identify the tissues and Cells
3. Preparation of Slides and Micrometry
4. Examination of prokaryotic and eukaryotic cell
5. Visualization of animal and plant cell by methylene blue
6. Visualization of nuclear fraction by aceto carmine stain
7. Staining and visualization of mitochondria by Janus green stain

II GROUP EXPERIMENT

8. Identification of different stages of meiosis in onion bulb
9. Identification of different stages of mitosis in onion root tip

III SPOTTERS

10. a) **Cells**: Nerve, plant and Animal cell
    b) **Organelles**: Mitochondria, Chloroplast, Endoplasmic reticulum,
    c) **Mitosis stages**–Prophase, Anaphase, Metaphase, Telophase

TEXT BOOKS

1. Rickwood,D and J.R.Harris cell Biology: Essential Techniques, Johnwikey1996.
2. Davis, J.M. Basic Cell culture: A practical approach, IRL 1994.
3. Ganesh M.K. and Shivashankara A.R. 2012. Laboratory Manual for Practical Biochemistry Jaypee publications, 2ndEdn.

REFERENCE BOOKS

1. Essential practical handbook of Cell biology ,Genetics and Microbiology -A Practical manual- Debarati Das Academic publishers, ISBN, 9789383420599, 1st Edition 2017
2.Cell biology Practical, Dr.Venugupta ISBN 8193651219, Prestige publisher, 1stJan2018.3. Cell and Molecular biology, DeRobertis, 8th edition, 1st June, 1987

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SEMESTER – III

SECOND YEAR	CORE COURSE - III	CREDIT - 5
SEMESTER - III	BIOCHEMICAL TECHNIQUES	25BCCC3

COURSE OUTCOMES

The objectives of this course are to

- Introduce the basic principles, types and applications of various sedimentation technique.
- Provide an understanding of the underlying principles of chromatographic techniques
- Demonstrate experimental skills in various electrophoretic techniques.
- Appraise the use of colorimetric and spectroscopic techniques in biology
- Impart knowledge about the measurement of radioactivity and safety aspects of radioactive isotopes.

COURSE SPECIFIC OUTCOMES

- Describe types of rotors and identify the centrifugation technique for the separation of biomolecules.
- Demonstrate the principles, operational procedure and applications of planar and column chromatography.
- Specify the factors and explain the separation of DNA and protein using electrophoretic technique.
- State Beer's Law and illustrate the instrumentation and uses of colorimeter and spectrophotometer.
- Enumerate various methods of measurement of radioactivity and safety aspects of radioactive isotopes.

Hours 15

UNIT - I

Centrifugation - Basic principles, RCF, Sedimentation coefficient, Svedberg constant. Types of rotors. Preparative centrifugation- differential and density gradient centrifugation, Rate zonal and Isopycnic techniques, construction, working and applications of analytical Ultracentrifuge - Determination of molecular weight (Derivation excluded).

Hours 15

UNIT II

Chromatography - adsorption, partition. Principle, instrumentation and applications of paper chromatography, thin layer chromatography, ion-exchange chromatography, gel permeation chromatography and affinity chromatography.

Hours 15

UNIT III

Electrophoresis – General principles, factors affecting electrophoretic mobility. Tiselius moving boundary electrophoresis. Electrophoresis with paper and starch. Principle, instrumentationandapplicationsofagarosegelelectrophoresisandSDS-PAGE.

UNIT IV **Hours 15**

Basics of Electromagnetic radiations- Energy, wavelength, wavenumber and frequency. Absorption and emission spectra, Lambert – Beer Law, Light absorption and transmittance.

Colorimetry - Principle, instrumentation and applications. Visible and UV spectrophotometry - Principle, instrumentation and applications –enzyme assay, structural studies of protein sand nucleic acids.

UNIT V **Hours 15**

Radioactivity - Types of Radioactive decay, half-life, units of radioactivity, Detection and measurement of radioactivity - Methods based upon ionization -Geiger Muller Counter. Methods based upon excitation - Solid & Liquid scintillation counters. Autoradiography. Biological applications and safety aspects of radioisotopes.

REFERENCE BOOKS

1. Avinash Upadhyay, Kakoli Upadhyay and Nirmalendu Nath, (2002). Biophysical Chemistry, Principles and Techniques, 3rd edition, Himalaya Publishing House.
2. Veerakumari L (2009). Bioinstrumentation, 1stedition, MJP Publishers.
3. Keith Wilson and John Walker, (2000). Practical Biochemistry - Principles and techniques, Cambridge University Press, 4th edition.
4. Terrance G. Cooper (1977). The tools of Biochemistry, John Wiley & Sons, Singapore.
5. Gurumani (2011). Research Methodology for Biological Sciences, 1stedition, MJP Publishers.
6. Saroj Dua and Neera Garg (2010). Biochemical Methods of Analysis, 1stedition, Narosa Publishing house.

WEB RESOURCES

1. https://www.britannica.com/science/chromatography.
2. https://www.youtube.com/watch?v=xgxFBQZYXlE.
3. https://www.youtube.com/watch?v=7onjVBsQwQ8.

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SECOND YEAR	CORE PRACTICAL - III	CREDIT - 4
SEMESTER - III	BIOCHEMICAL TECHNIQUES (P)	25BCCP3

COURSE OUTCOMES

The objectives of this course are to:

- Acquaint the students with colorimetric estimations of biomolecules.
- Equip skills on various separation techniques.
- Impart knowledge about the estimation of minerals and vitamins.

COURSE SPECIFIC OUTCOMES

- Estimate the amount of biomolecules by colorimetric method.
- Quantify the amount of minerals by colorimetric method.
- Separate and identify sugars, lipids and amino acids by chromatography.
- Operate centrifuge for the separation of serum and plasma.
- Demonstrate the separation of proteins electrophoretically.

I COLORIMETRY

HOURS : 60

1. Estimation of amino acid by Ninhydrin method.
2. Estimation of protein by Biuret method.
3. Estimation of DNA by Diphenylamine method.
4. Estimation of RNA by Orcinol method.
5. Estimation of Phosphorus by Fiske and Subbarow method.

II CHROMATOGRAPHY

1. Separation and identification of sugars and amino acids by paper chromatography.
2. Separation and identification of amino acids and lipids by thin layer chromatography.

III DEMONSTRATION

1. Separation of serum and plasma from blood by centrifugation.
2. Separation of serum proteins by SDS-PAGE.

REFERENCE BOOKS

- Jayaraman J (2015). Laboratory Manual in Biochemistry New Age International Limited Fifth edition.
- Sadasivam S and Manickam A (2018). Biochemical Methods New Age International Pvt. Ltd publishers 3rd edition.
- Keith Wilson and John Walker (2010). Principles and techniques of Practical Biochemistry Cambridge University Press, Seventh edition.
- Sawhney SK and Randhir Singh (2005). Introductory Practical Biochemistry. Alpha Science International, Ltd 2nd edition.
- David T Plummer (2001). An Introduction to Practical Biochemistry, 3rd edition, Tata McGraw- Hill publishing company limited.
- Alan H Gowenlock (1988). Varley's Practical Clinical Biochemistry by, published by CBS Publishers and distributors, India Sixth Edition.

WEB RESOURCES

- https://www.pdfdrive.com/biochemistry-books.html.

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SECOND YEAR	NON MAJOR ELECTIVE COURSE - I	CREDIT : 2
SEMESTER -III	MEDICINAL DIET	25NMBC1A

COURSE OUTCOMES

The main objectives of this course are to

- Provide basic knowledge about diet.
- Understand of diet modification for GI diseases.
- Plan a diet for liver diseases.
- Prepare diet chart for Infectious diseases.
- Plan a diet for Diabetes, Renal and Cardio-vascular diseases.

COURSE SPECIFIC OUTCOMES

- Possess basic knowledge about diet.
- Sketch diet plan for GI diseases.
- Sketch diet plan for liver diseases.
- Sketch a diet plan for Infectious diseases.
- Prepare diet chart for Diabetes Renaland Cardio-vascular diseases.

Hours 6

UNIT – I

Principles of Therapeutic Diet: Definitions of Normal diet, Therapeutic diet, soft Diet and Liquid diet. Objectives of Diet Therapy. Advantages of using normal diet as the basis for Therapeutic diet. Normal Diet-therapeutic modification of normal diet.

Hours 6

UNIT – II

Diet modification in Gastrointestinal diseases: Peptic ulcer, Diarrhea, Lactose intolerance, Constipation and Malabsorption syndrome.

Hours 6

UNIT – III

Diet Modification in liver and gall bladder in diseases: Etiology, symptoms and dietary treatment in jaundice, hepatitis, cirrhosis of liver and hepatic coma.

Hours 6

UNIT – IV

Diet Modification in Infectious Diseases: Fevers, Typhoid, Tuberculosis and Viral Hepatitis. Dietary modifications in Tuberculosis.

UNIT – V **Hours 6**

Module V: Diet Modification in Diabetes, Renal and Cardio-vascular Diseases-Diabetes, acute & chronic glomerulonephritis, nephrosis, renal failure, kidney stone and Hypertension. 6 Hrs

REFERENCE BOOKS

1. M.Raheena Begum, AText Book of Foods, Nutrition and Dietetics, Sterling Publishers Pvt. Ltd.
2. M.V.Raja Gopal, Sumati.R.,Mudambi, Fundamentals of foods and Nutrition, Wiley Eastern Limited, Year1990.
3. William S.R Nutrition and Diet Therapy, 1985, 5thedition, Mosly Co. St. Louis.
Reference books
1. Rodwell Williams Nutrition and Diet Therapy, 1985, the C.V Mosly St. Louis.
2. M.V.Krause & M.A.Mohan, Food Nutrition and Diet Therapy, 1992 by W.B Saunders Company,
Philadelphia, London.
3. Davidson and Passmore, Human Methods and Diabetics, 1976 the English Language Book Society and
Churchill.

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SECOND YEAR	NON MAJOR ELECTIVE COURSE - I	CREDIT : 2
SEMESTER -III	HEALTH AND NUTRITION	25NMBC1B

COURSE OUTCOMES

The main objectives of this course are to

- The prime focus is to provide the students with a basic understanding of the relationship between food, nutrition and health.
- It is imperative that focus should be on realistic issues faced by people with respect to nourishment at all stages of life.
- Unhealthy eating habits particularly the shift from fresh food consumption to packaged foods with added salts and preservatives have contributed to the obesity epidemic in nearly all parts of the world.

COURSE SPECIFIC OUTCOMES

Upon the completion of the course, students will be able to:

- Have a better understanding of the association of food and nutrition in promoting healthy living.
- Think more holistically about the relationship between nutrition science, social and health issues.
- Move on to do post-graduation studies and can apply for jobs as food safety officers, food analysts, food inspectors, food safety commissioners or controllers for jobs in organizations like FSSAI.
- Specialize in various fields of nutrition.

Hours 6

UNIT I:

Health: Definition and concept of health, Major nutritional Deficiency diseases- (kwashiorkor and marasmus), Deficiency disorders, their causes, symptoms, treatment, prevention and government programmes, if any. Life style related diseases- hypertension, diabetes mellitus, Atherosclerosis and obesity- their causes and prevention through dietary and lifestyle modifications, Social health problems- smoking, alcoholism, drug dependence and Common ailments- cold, cough, and fevers, their causes and treatment.

Hours 6

UNIT II:

Basic concept of food and nutrition: Food Components and food-nutrients, Concept of a balanced diet, nutrient needs and dietary pattern for various groups- adults, pregnant and nursing mothers, infants, school children, adolescents and elderly. Food Pyramid, Nutritional anthropometry- BMI, waist- to-hip ratio, skin-fold test and bioelectrical impedance; interpretation of these measurements.

UNIT III: **Hours 6**

Nutritional Biochemistry: Carbohydrates, Lipids, Proteins, their dietary source and role Vitamins- their dietary source and importance Minerals- their biological functions. Dietary Fibres - Definition, their dietary source and nutritional importance. Elementary idea of Probiotics, Prebiotics, Organic Food.

UNIT IV: **Hours 6**

Food hygiene: Food and Water borne infections; Bacterial infection: Cholera, typhoid fever, dysentery; Viral infection: Hepatitis, Poliomyelitis; Protozoan infection: amoebiasis, giardiasis; Parasitic infection: taeniasis and ascariasis their transmission, causative agent, sources of infection, symptoms and prevention; Brief account of food spoilage: Causes of food spoilage and their preventive measures.

UNIT – V **Hours 6**

Interrelationship between nutrition/nutrients & Health: Functions, dietary sources and clinical manifestations of deficiency/ excess of the following nutrients: Carbohydrates, lipids and proteins; Fat soluble vitamins A&D; Water soluble vitamins – thiamine, riboflavin and vitamin C; Minerals - calcium, Iodine and iron. Conservation of Nutrients: Minimizing nutrient losses during pre-preparation and preparation of cereals, pulses, fruits and vegetables, milk, oil, egg, meat, fish & poultry

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SECOND YEAR	ACTIVITY BASED COURSE	CREDIT : 1
SEMESTER - III	HEALTH & WELLNESS	25HWUGC3

COURSE OUTCOMES

The main objectives of this course are to

- Focuses on teaching the elements of physical, mental, emotional, social, intellectual, environmental well-being which are essential for overall development of an individual.
- The course also addresses the dangers of substance abuse and online risks to promote emotional and mental health.

COURSE SPECIFIC OUTCOMES

Upon completion of the Health & Wellness course, students will be able to:

- Demonstrate proficiency in sports training and physical fitness practices.
- Improve their mental and emotional well-being, fostering a positive outlook on health and life.
- Develop competence and commitment as professionals in the field of health and wellness.
- Awareness on drug addiction and its ill effects.

UNIT – I

Introduction to Health & Wellness: Definition of health- WHO definition, Importance of health in everyday life, Components of health- physical, social, mental, spiritual and its relevance, Concept of wellness, Mental Health & wellness, Determinants of health behaviours, Using the mass media for health promotion.

UNIT – II

Nutrition and Weight Management: Meaning, Definition and Importance of Nutrition, Food and effect of malnutrition on health, Mid-Day Meal. Basic Nutrients (Protein, Carbohydrate, Fat, Vitamins, Mineral & Water), Phytonutrients, Fibrous Food. Diet, Balance Diet, Athletic Diet, Factors affecting Diet. Obesity – Concept, Problems, Causes, Prevention, Assessment and procedure of weight Management.

UNIT – III

Hygiene, personal hygiene, mental hygiene & community Hygiene: Meaning, Concept and types of Hygiene. Importance of Hygiene for healthy life, desirable hygienic habits and Importance of rest, sleep & exercise. Personal Hygiene: - Care of Skin, Eye, Teeth, Ear, Nail, Nose and Hair. Mental Hygiene, its importance and its practice procedure.

UNIT – IV

Health Problem in India: Causes, Prevention and Control of Communicable Diseases: Malaria, Dengue, Corona virus. Causes, Prevention and Control Non-Communicable Diseases: Thalassemia, Asthma, Arthritis. Postural Deformities: Causes and corrective exercise of Kyphosis, Lordosis, Scoliosis, Knock Knees, Bow leg and Flat foot. Problems associated with postural deformities. Life Style Disease (Diabetes, Hypertensions, Stroke) and Stress Management.

UNIT – V

Mind – body and well-being: Mind- Body connection in health- concept and relation Implications of mind-body connections. Wellbeing- why it matters? Digital wellbeing Understanding health beliefs, and perspectives of indigenous people pertaining to Assam and North East India. Promoting Human strengths and life enhancement: Classification of human strengths and virtues; cultivating inner strengths: Hope and optimism.

REFERENCE BOOKS

- Carr A (2004). Positive Psychology: The science of happiness and human strength. UK: Routledge.
- Hick JW (2005). Fifty signs of Mental Health. A Guide to understanding mental health. Yale University Press.
- Snyder CR and Lopez SJ (2007). Positive psychology: The scientific and practical explorations of human strengths. Thousand Oaks, CA: Sage.
- Edlin G and Golanty E (2019). Health & Wellness, 13th ed. United States of America: Jones & Bartlett Learning, Suggested References.
- E. Hardman and D. J. Stensel, D. J., Physical Activity and Health: The Evidence Explained, 2nd ed. London and New York: Routledge, Taylor & Francis Group, 2009.
- K. L. Harkness and E. P. Hayden, Eds., The Oxford Handbook of Stress and Mental Health. New York: Oxford University Press, 2020.
- Human Kinetics, Health and Wellness for Life. Health Textbooks. United States of America: Human Kinetics, Inc., 2010.
- Thakur, S. Krira Chikitsha, Paschimbanga Rajya Pustak Parsad.
- Hanlon, John J (2003). "Principles of Public Health Administration".Nutrition Encyclopaedia, edited by Delores C.S. James, The Gale Group, Inc.
- Ghosh, B.N. A Treaties of Hygiene and Public Health, Scientific Publishing Co., Kolkata.
- Boyd-Eaton S et al (1989) The Stone Age Health Programme: Diet and Exercise as Nature Intended. Angus and Robertson.

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SEMESTER – IV

SECOND YEAR	CORE COURSE - IV	CREDIT : 5
SEMESTER - IV	HUMAN PHYSIOLOGY	25BCCC4

COURSE OUTCOMES

The main objectives of this course are to

- Aid in understanding the physiology of respiratory and circulatory systems.
- Explain the structure and physiology of the nervous and muscular system.
- Explicate the functions of digestive and excretory system of the body.
- Impart knowledge about the process of reproduction.
- Emphasize the importance of various endocrine factors that regulate metabolism, growth, homeostasis and reproduction.

COURSE SPECIFIC OUTCOMES

- Explain the exchange of gases, design of blood vessels and cardiac cycle.
- Summarize the events in transmission of nerve impulse sand mechanism of muscle contraction.
- Elaborate the structure and functions of digestive system, structure of nephron and mechanism of urine formation and role of kidney in maintenance of pH.
- Describe the process of Oogenesis, Spermatogenesis, Fertilization, and Parturition.
- Understand the role of different hormones that regulate metabolism, growth, glucose homeostasis and reproductive function.

Hours:15

UNIT - I

Respiratory System: Overview of respiratory system, Types of respiration, Transport of respiratory gases, Exchange of respiratory gases in lungs and tissues –Chloride Shift & Bohr's effect, Lung surfactant. Circulatory System-Structure and functions of the Heart. Arterial and venous system, Cardiac cycle, Pacemaker, Blood pressure and Factors affecting blood pressure.

Hours:15

UNIT - II

Nervous system: Structure of neuron, synaptic transmission, reflex action, neurotransmission- Resting membrane and Action potential. neuro transmitters- acetyl choline, Noradrenaline, Dopamine, Serotonin, Histamine, GABA, Substance P. Muscular system-structure and types of muscles - skeletal, smooth and cardiac muscles, muscle proteins- types and functions, mechanism of muscle contraction.

Hours:15

UNIT - III

Digestive system: composition, functions of saliva, gastric pancreatic intestine and bile secretions, structure of digestive system, Digestion, absorption of carbohydrates, lipids, proteins. Excretory system – Structure of nephron, mechanism of urine formation, Concentration and acidification of Urine. Role of kidneys in the maintenance of acid base balance.

UNIT - IV **Hours:15**

Reproductive system: Oogenesis, spermatogenesis, capacitation and transport of sperm- blood testis barrier. Fertilization, early development, Implantation, Placentation and Parturition.

UNIT - V **Hours:15**

Endocrinology: Classification of hormones, endocrine glands and their secretions, structure and functions of Insulin, thyroxine. Steroid hormones- Corticosteroids, Sex hormones – testosterone and estrogen, menstrual cycle.

REFERENCE BOOKS

- Sembulingam K and Prema Sembulingam (2016). Essentials of Medical Physiology, 7th edition, Jaypee Brothers Medical Publishers (P) Ltd.
- Chatterjee CC (1988). Human Physiology – Vol I & II,1st edition, Medical Allied Agency.
- Mariakuttikan and Arumugam (2017). Animal Physiology, Saras publication.
- Chatterjee MN and Rana shinde (2007). Text book of medical biochemistry physiology- 7th edition, Jaypee brothers- medical publishers.
- Meyer, Meyer and Meij (2002). Human Physiology, 3rd edition, A. I. T. B. S. Publishers.
- Guyton and Hall (2010). Textbook of Medical Physiology, 12th edition, Saunders Publishers.
- Guytonand Hall (2011). Textbook of Medical Physiology, 12th edition, W. B. Saunders Company.
- Elaine N. Marieb (1995). Human anatomy and physiology, 3rd edition, Benjamin/Cummings (a Pearson education company).

WEB RESOURCES

- https://www.youtube.com/watch?v=6qnSsV2syUE
- https://www.youtube.com/watch?v=9_h0ZXx1lFw
- https://slideplayer.com/slide/9431799/

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SECOND YEAR	CORE PRACTICAL - IV	CREDIT : 4
SEMESTER -VI	HUMAN PHYSIOLOGY (P)	25BCCP4

COURSE OUTCOMES

The main objectives of this course are to

- Gain knowledge on the counting of total RBC, WBC and Platelets.
- To understand the Eosinophil count.

COURSE SPECIFIC OUTCOMES

- To help the Determination of Haemoglobin content, clotting time, Prothrombin time, ESR.

HUMAN PHYSIOLOGY METHODS

HOURS: 60

1. Collection, storage of blood and anticoagulants
2. Blood Grouping
3. Total RBC count
4. Total WBC count
5. Total platelet count
6. Differential WBC count
7. Absolute Eosinophil count
8. Determination of Haemoglobin content
9. Determination of clotting time.
10. Determination of Prothrombin time
11. Determination of ESR.

REFERENCE BOOKS

- Hematology Laboratory Manual (English, Paperback, Alabdallat Nessrin Ghazi Mohammad).
- Hematology: Basic Principles and Practice 7th Edition by Ronald Hoffman and Edward J Benz and Leslie E Silberstein and Helen Heslop and Jeffrey Weitz and John Anastasi, Elsevier Science.
- Basic Medical Laboratory Techniques – Estridge, Reynold and Walter- 4th edition.
- Medical Lab Technology- Kanai L Mukherjee.
- Medical Lab Technology – Ramnik Sood.
- Pattabiraman TN Laboratory manual in Biochemistry –3rd edition.
- An Introduction to Practical Biochemistry - David T. Plummer Third Edition, Tata Mc Graw Hill.
- Medical laboratory Technology Volume I, II & III - KL Mukherjee.

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SECOND YEAR	NON MAJOR ELECTIVE COURSE - II	CREDIT : 2
SEMESTER - III	FIRST AID AND HOME NURSING	25BCNM2A

COURSE OUTCOMES

The main objectives of this course are to

- Provide knowledge on the basics of first aid.
- Perform first aid during various respiratory issues.
- Demonstrate the first aid to treat injuries.
- Learn the first aid techniques to be given during emergency.
- Familiarize the first aid during poisoning.

COURSE SPECIFIC OUTCOMES

- Discuss on the rules of first aid, dealing during emergency and first aid techniques.
- Understand the first aid techniques to be given during different types of respiratory problems.
- Provide first aid for injuries, shocks and bone injury.
- Detail on the first aid to be given for unconsciousness, stroke, fits and convulsions.
- Gain expertise in giving first aid for insect bites and chemical poisoning.

UNIT - I **Hours: 6**

Aims and important rules of first aid, dealing with emergency, types and content of a first aid kit. First aid technique – Dressing and Bandages, fast evacuation technique, transport techniques.

UNIT- II **Hours: 6**

Basics of Respiration – CPR, first aid during difficult breathing, drowning, choking, strangulation and hanging, swelling within the throat, suffocation by smoke or gases and asthma.

UNIT- III **Hours: 6**

Common medical aid- first aid for wounds, cuts, head, chest, abdominal injuries, shocks, burns, amputations, fractures, dislocation of bones.

UNIT- IV **Hours: 6**

Home nursing - definition and roles; sick room, observation, infection, diet. Unconscious patient - causes, history and observe. Elementary child care - immunization of new born and recognition of childhood disease. Environmental sanitation- clean house, safe drinking water.

THIRD YEAR	NON MAJOR ELECTIVE COURSE - II	CREDIT : 2
SEMESTER - IV	LIFE STYLE DISEASES	25BCNM2B

COURSE OUTCOMES

The objectives of this course are to

- Create awareness on lifestyle diseases among adolescents.
- List out the lifestyle diseases.
- Explain the common lifestyle diseases and their prevention.
- Acquaint the disorders associated with women's health.
- Impart life skills so as to prevent life style diseases.

COURSE SPECIFIC OUTCOMES

- Define life style diseases and describe the contributing factors
- Enumerate the top lifestyle diseases and its impact on life.
- Elaborate the treatment and prevention measures of common lifestyle diseases.
- Highlight the life style diseases that affects the women's health
- Illustrate the various measures for prevention of life style diseases

UNIT - I **Hours: 6**

Lifestyle diseases: Definition, Factors contributing to lifestyle diseases – Physical inactivity, Poor food habits, disturbed biological clock, sleep deprivation.

UNIT - II **Hours: 6**

Top lifestyle diseases, Impact of Lifestyle diseases on family, society and economy of country.

UNIT - III **Hours: 6**

Causes, symptoms, types, preventive measures and treatment of Obesity, cardiovascular diseases, diabetes and cancer.

UNIT - IV **Hours: 6**

Women's lifestyle diseases: Polycystic Ovarian Disease, Infertility, Breast and cervical cancer and Osteoporosis.

UNIT - V **Hours: 6**

Prevention of lifestyle diseases: Balanced diet, sufficient intake of water, physical activity, sleep-wake cycle, stress management and meditation.

REFERENCE BOOKS

- James MR (2013). Life style medicine, 2nd edition, CRC press,
- Akira Miyazaki (2008). New frontiers in lifestyle-related disease, Springer.
- Steyn K, life style and related risk factors for chronic diseases.
- Willett WC, Prevention of chronic disease by means of diet and lifestyle.
- Kumar M & R. Kumar. Guide to prevention of life style diseases. Deep & Deep publications.

WEB RESOURCES

- https://youtu.be/jDdL2bMQXfE.
- https://youtu.be/7WnpSB14nDM.

https://youtu.be/ollz9MqtW-U.

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Hours: 6

UNIT- V

Simple nursing: bedding, temperature of foreign body - ear, nose and throat. Bleeding of nose and ear - insects' bites, sitting poisons. Prenatal and postnatal care: introduction, medical complications. Preventive orthopaedics: introduction and Prevention of orthopaedics disabilities

REFERENCE BOOKS

- Gauri Goel, Kumkum Rajput, Manjul Mungali First aid and health 1SBN-978 92208-19-5.
- Indian First Aid Mannual-https://www.indianredcross.org/publications/FA-manual.pdf
- Red Cross First Aid/CPR/AED Instructor Manual.
- Mayilvaganan Natrajan R (1993). Principals of first aid and home nursing.
- Eveleen Harrison A (2018). Text book of home nursing.

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SEMESTER – V

THIRD YEAR	CORE COURSE - V	CREDIT : 5
SEMESTER - V	ENZYMES	25BCCC5

COURSE OUTCOMES

The main objectives of this course are to

- Provide fundamental knowledge on enzymes and their properties.
- Understand the mechanism of action of enzymes and the role of coenzymes in catalysis.
- Introduce the kinetics of enzymes and determine the Km and Vmax.
- Explain the effect of inhibitors on enzyme activity.
- Understand the role of enzymes in clinical diagnosis and industries.

COURSE SPECIFIC OUTCOMES

- Identify the major classes of enzymes, differentiate between a chemical catalyst and a biocatalyst and define the units of enzymes.
- Explain the mechanism of enzyme catalysis and the role of coenzymes in enzyme action.
- Illustrate the steady state kinetics, interpret MM plot and LB plot based on kinetics data, and determine Km and Vmax.
- Distinguish the types of inhibition along with its importance in biochemical reactions.
- Comprehend the various methods for production of immobilized enzymes and discuss the application of enzymes in clinical diagnosis and various industries.

Hours: 18

UNIT - I

Introduction to enzymes: Nomenclature and Classification based on IUB with examples, enzyme as catalyst - Activation energy, Enzyme specificity-absolute, Group, linkage and stereo specificities. Concept of Active site, Lock and key hypothesis and induced fit theory, Enzyme expression Units-IU, turnover number, katal and specific activity.

Hours: 18

UNIT - II

Mechanism of enzyme catalysis: Acid Base catalysis, covalent catalysis, electrostatic catalysis, metal ion catalysis, proximity and orientation effect. Coenzymes -Definition, types, co-enzymatic forms of vitamins- NAD/NADP, FAD, FMN, Coenzyme A TPP, PLP, lipoic acid and biotin. Multienzyme complexes - Pyruvate dehydrogenase complex. Isoenzyme with reference to LDH and CK.

Hours: 18

UNIT - III

Enzyme kinetics: Definition of kinetics, Factors affecting enzyme activity - temperature, pH, substrate and enzyme concentration, activators-cofactors, Derivation of Michaelis - Menton equation for unisubstrate reactions, Lineweaver - Burk plot, Eadie –Hof steep lot Significance of Km and V max and the IR determination using the plots.

UNIT - IV **Hours: 18**

Enzyme inhibition: Reversible and irreversible inhibition-types of reversible inhibitors, competitive, non-competitive, un-competitive inhibitors. Graphical representation by L-B plot, (Kinetic derivations not required), Determination of Km and V max in the presence and absence of inhibitors. Allosteric enzymes - Sigmoidal curve, positive and negative modulators.

UNIT - V **Hours: 18**

Applications of enzymes: Immobilized enzymes - methods of immobilization-adsorption, covalent bonding, crosslinking, encapsulation, entrapment and applications of immobilized enzymes. Biosensors – e.g. Glucose sensors. Industrial applications of enzymes – Food, textile and pharmaceutical industries, Clinical application of enzymes.

REFERENCE BOOKS

1. Sathyanarayana U and Chakrapani U (2013). Biochemistry, 4th edition, Elsevier India Pvt. Ltd., Books and Allied Pvt. Ltd.
2. Agarwal GR, Kiran Agarwal and Agarwal OP (2015). Textbook of Biochemistry (Physiologicalchemistry), 18th edition, Goel Publishing House.
3. Devasena T (2010). Enzymology, 1st edition, Oxford University Press.
4. Trevor Palmer (2008). Enzymes: Biochemistry, Biotechnology, Clinical Chemistry, 2nd edition, East West Press Pvt. Ltd.
5. David L Nelson and Michael M Cox, (2005). Principles of Biochemistry, 4th edition WH. Freeman and Company.
6. Voet D, Voet JG and Pratt, CW (2004). Principles of Biochemistry, 4th edition John Wiley & Sons, Inc.
7. Zubay GL *et.al.* (1995). Principles of Biochemistry, 1st edition, WmC. Brown Publishers.

WEB RESOURCES

1. www.biologydiscussion.com/notes/enzymesnotes.
2. https:// www.britannica.com/science/protein/The-mechanism-of-enzymatic-action.
3. https://www.youtube.com/watch?v=oVJ2LJxO6tU.

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THIRD YEAR	CORE COURSES - VI	CREDIT : 5
SEMESTER - V	MOLECULAR BIOLOGY	25BCCC6

COURSE OUTCOMES

The objectives of this course are to

- Provide insights into the central dogma of molecular biology and explain the mechanism of DNA replication.
- Elaborate the mechanism of transcription and reverse transcription.
- Highlight the characteristics of genetic code and describe the process of protein synthesis.
- Introduce the concept of regulation of gene expression in prokaryotes
- Familiarize the different types of mutations and explain the mechanism of DNA repair.

COURSE SPECIFIC OUTCOMES

- Illustrate the Central Dogma of molecular biology, explain the multiplication of DNA in the cell and describe the types and modes of replication.
- Elaborate the mechanism of transcribing DNA into RNA, discuss the formation of different types of RNA.
- Decipher the genetic code and summarize the process of translation.
- Comprehend the principles of gene expression and explain the concept of operon in prokaryotes.
- Distinguish the types of mutations and explain the various mechanisms of DNA repair.

UNIT - I **Hours: 18**

Central Dogma of molecular Biology, DNA as the unit of inheritance. Experimental evidences by Griffith's transforming principle, Avery, McLeod and McCarthy's experiment, and Hershey and Chase Experiment. Replication in prokaryotes: Modes of replication, Meselson and Stahl's experimental proof for semiconservative replication. Mechanism of Replication – Initiation, events at Ori C, Elongation – replication fork, semi discontinuous replication, Okazaki fragments, and termination. Bidirectional replication, Inhibitors of replication. Models of replication-theta, rolling circle and D loop model.

UNIT - II **Hours: 18**

Transcription - Mechanism of transcription: DNA dependent RNA polymerase(s), recognition, binding and initiation sites, TATA/ Pribnow box, elongation and termination. Post-transcriptional modifications; inhibitors of transcription. RNA splicing and processing of mRNA, tRNA and rRNA. Reverse transcription.

UNIT - III **Hours: 18**

Genetic Code and its characteristics, Wobble hypothesis. Translation: Adaptor role of tRNA, Activation of amino acids, Initiation, elongation and termination of protein synthesis, post-translational modifications and inhibitors of protein synthesis.

UNIT - IV **Hours: 18**

Regulation of Gene Expression in Prokaryotes – Principles of gene regulation, negative and positive regulation, concept of operons, regulatory proteins, activators, repressors, regulation of lac operon and trp operon.

UNIT - V **Hours: 18**

Mutation: Types- Nutritional, Lethal, Conditional mutants. Missense mutation and other point mutations. Spontaneous mutations; chemical and radiation – induced mutations. DNA repair: Direct repair, Photo reactivation, Excision repair, Mismatch repair, Recombination repair and SOS repair.

REFERENCE BOOKS

- Veer Bala Rastogi (2008). Fundamentals of Molecular Biology, 1st edition, Ane books India.
- David Friefelder (1987). Molecular Biology, 2nd edition, Narosa Publishing House.
- Verma PS and Agarwal VK (2013). Cell biology, Genetics, Molecular Biology, Evolution and Ecology,1st edition, S. Chand & Company Pvt. Ltd.
- Karp G (2010). Cell and Molecular Biology: Concepts and Experiments, 6th edition, John Wiley & Sons. Inc.
- De Robertis EDP and De Robertis EMF (2010). Cell and Molecular Biology, 8th edition, Lippincott Williams and Wilkins, Philadelphia.
- James D. Watson (2013). Molecular Biology of the Gene 7th edition, Benjamin Cummings.
- George M Malacinski (1992). Freifelder's Essentials of Molecular Biology, 4th edition, Narosa publishing House.

WEB RESOURCES

- www.mednotes.net/notes/biology
- https://www.onlinebiologynotes.com/repair-mechanism-of mutation/
- https://teachmephysiology.com/biochemistry/protein-synthesis/dna-translation/

BOS CHAIRMAN

THIRD YEAR	CORE COURSE - VII	CREDIT : 5
SEMESTER - V	INTERMEDIARY METABOLISM	25BCCC7

COURSE OUTCOMES

The main objectives of this course are to

- Review the basic concepts of free energy transformation and describe biological oxidation.
- Illustrate the pathways of carbohydrate metabolism.
- Explain the pathways of oxidation and biosynthesis of lipids.
- Detail the catabolism of amino acids and synthesis of specialized products from amino acids.
- Acquaint the metabolism of nucleic acids and its regulation.

COURSE SPECIFIC OUTCOMES

- State the concepts of bioenergetics and illustrate the mechanism of flow of electrons and the production of ATP.
- Elaborate the biochemical reactions and integration of pathways of carbohydrate metabolism.
- Sketch the oxidation and biosynthesis of fatty acids, phospholipids, triglycerides and cholesterol with suitable examples.
- Explain catabolism of amino acids, synthesis of non-essential amino acids and specialized products from amino acids.
- Describe the metabolism of nucleic acids with necessary illustrations and its regulation.

UNIT - I **Hours: 18**

Bioenergetics - High energy compounds: Role of high energy compounds, free energy hydrolysis of ATP and other organophosphates, ATP-ADP cycle. Biological Oxidation: Electron transport chain - its organization and function. Inhibitors of ETC. Oxidative phosphorylation, P/Oratio, Peter Mitchell's chemiosmotic hypothesis. Mechanism of ATP synthesis, uncouplers of oxidative phosphorylation, substrate level phosphorylation with examples.

UNIT - II **Hours: 18**

Metabolism of carbohydrates -Glycolysis, TCA Cycle, Amphibolic nature and integrating role of TCA cycle. Anaplerosis, Pentose Phosphate Pathway (HMP shunt), Gluconeogenesis, Glycogenesis, Glycogenolysis and its regulation, glyoxylate cycle, Entner-Duodoroff pathway and Coricycle.

UNIT - III **Hours: 18**

Metabolism of lipids -Oxidation of fatty acids - α, β and ω -oxidation of saturated fatty acids, Oxidation of fatty acids with odd number of carbon atoms and unsaturated fatty acids, Ketogenesis, Biosynthesis of saturated fatty acids and unsaturated fatty acids, Biosynthesis and degradation of triglycerides, phospholipids and cholesterol.

UNIT - IV **Hours: 18**

Metabolism of amino acid- Metabolic nitrogen pool, Catabolism of amino acid: Oxidative deamination, non – oxidative deamination, transamination and decarboxylation, Biogenic amines, Urea cycle and its regulation.

UNIT – V **Hours: 18**

Metabolism of nucleotides - Biosynthesis of purines and pyrimidines, - denovo synthesis and salvage pathways, Degradation of purines and pyrimidines, Conversion of ribonucleotide to deoxyribonucleotide.

REFERENCE BOOKS

- Sathyanarayana U and Chakrapani U (2015). Biochemistry, 4th Elsevier India Pvt. Ltd.
- Chatterjea MN and Rana Shinde (2002). Textbook of Medical Biochemistry, 5th edition Jaypee Brothers Medical Publishers Pvt. Ltd.
- David L Nelson, Michael M Cox (2008). Lehninger Principles of Biochemistry, 5th edition, W. H. Freeman and Company.
- Robert K Murray, Daryl K. Granner and Victor W Rodwell (2006). Harper's Illustrated Biochemistry, 27th edition, McGraw Hill Publishers.
- Voet D, Voet J. G and Pratt CW (2010). Principles of Biochemistry, Fourth edition, John Wiley & Sons, Inc.
- Geoffrey L Zubay, William W Parson and Dennis E Vance (1995). Principles of Biochemistry, 2nd Edition, Wm. C. Brown Publishers.
- Garret RH and Grisham CM (2005). Biochemistry, 3rd Edition. Thomson Learning INC.

WEB RESOURCES

- https://nptel.ac.in/courses/104/105/104105102/.
- http://www.nptelvideos.in/2012/11/biochemistry-i.html.
- https://www.saddleback.edu/faculty/jzoval/mypptlectures/ch15_metabolism/lecture_notes_ ch15_metabolism_current-v2.0.pdf.

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THIRD YEAR	CORE PRACTICAL - V	CREDIT : 4
SEMESTER -V	ENZYME AND MOLECULAR BIOLOGY (P)	25BCCP5

COURSE OUTCOMES

- To gain the knowledge regarding different analytical techniques for Biomolecules
- To understand the principle and techniques for isolation procedure

COURSE SPECIFIC OUTCOMES

- To estimate and analyze DNA and RNA using standard methods in body fluids.

EXPERIMENTS

HOURS: 60

1. Isolation of enzymes from natural sources.
2. Determination of specific activity of salivary amylase.
3. Determination of effect of temperature on salivary amylase.
4. Determination of the effect of pH on salivary amylase.
5. Determination of KM and Vmax using Line weaver-Burk plot for salivary amylase.
6. Determination of activators for salivary amylase.
7. Determination of inhibitors for salivary amylase.
8. Isolation of DNA/RNA.
9. Estimation of DNA by Diphenylamine method.
10. Estimation of RNA by Orcinol method.

REFERENCE BOOKS

- Price and Stevens 1989, Fundamentals of Enzymology, Oxford Bioscience publications 2nd Ed, New york.
- Satyanarayana U (2019). Fundamentals of Biochemistry, Allied & Books Pvt. Ltd, Calcutta.
- Jain JL (2005). Fundamentals of biochemistry, 6th Edition, S. Chand Publishers, New Delhi.
- Voet D and Voet JG. (1990). Biochemistry, 4th edition, John Wiley & SonsInc, Publishers, New York White, 1959.
- Principles of Biochemistry, 3rd edition, McGraw Hill Book Co., Publishers, New York.
- Lehninger (2004). Principles of Biochemistry: D.L. Nelson, Michael M. Cox, International
- Edition, CBS Publishers, 4th Ed.
- Stryer WH (1995). Biochemistry Freeman &Co., Scientific Research an academic Publisher 4th Ed., New York.
- Marangoni (2002). Enzyme kinetics. A modern approach, 1st edition, John Wiley Publishers, United Kingdom.
- Dixon M and Webb JF (1979). Enzymes, 2nd edition, Longman Publishers, London. Stryer.

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THIRD YEAR	MAJOR BASED ELECTIVE - I	CREDIT : 4
SEMESTER - V	NUTRITIONAL BIOCHEMISTRY	25BCMB1A

COURSE OUTCOMES

The objectives of this course are to

- Create awareness about the role of nutrients in maintaining proper health.
- Understand the nutritional significance of carbohydrates, lipids and proteins.
- Understand the importance of a balanced diet.
- Study the effect of additives, emulsifiers, flavour enhancing substances in food.
- Study the significance of nutraceuticals.

COURSE SPECIFIC OUTCOMES

- Cognizance of basic food groups viz. Carbohydrates, proteins and lipids and their nutritional aspects as well as calorific value.
- Identify and explain nutrients in foods and the specific functions in maintaining health.
- Classify the food groups and its significance.
- Understand the effect of food additives.
- Describe the importance of nutraceuticals and pigments.

UNIT - I **Hours: 12**

Concepts of food and nutrition. Basic food groups-energy yielding, body building and functional foods. Modules of energy. Calorific and nutritive value of foods. Measurement of Calories by bomb calorimeter. Basal metabolic rate (BMR)- definition, determination of BMR and factors affecting BMR. Respiratory quotient (RQ) of nutrients and factors affecting the RQ. SDA-definition and determination- Anthropometric measurement and indices – Height, Weight, chest and waist circumference BMI.

UNIT - II **Hours: 12**

Physiological role and nutritional significance of carbohydrates, lipids and protein. Evaluation of proteins by nitrogen balance method- Biological value of proteins- Digestibility coefficient, Protein Energy Ratio and Net Protein Utilization. Protein energy malnutrition – Kwashiorkor and Marasmus, Obesity-Types and preventive measures.

UNIT - III **Hours: 12**

Balanced diet, example of low and high cost balanced diet- for infants, children, adolescents, adults and elderly people. ICMR classification of five food groups and its significance food pyramid. Junk foods- definition and its adverse effects.

Hours: 12

UNIT - IV

Food additives: Structure, chemistry, function and application of preservatives, emulsifying agents, buffering agents, stabilizing agents, natural and artificial sweeteners, bleaching, starch modifiers, antimicrobials, food emulsions, fat replacers, viscosity agents, gelling agents and maturing agents. Food colours, flavours, anti-caking agent, antioxidants. Safety assessment of food additives.

Hours: 12

UNIT - V

Nutraceuticals and Functional Foods: Definition, properties and function of Nutraceuticals, food Supplements, dietary supplements prebiotics and probiotics, and functional Foods. Food as medicine. Natural pigments from plants– carotenoids, anthocyanins and its benefits.

REFERENCE BOOKS

- Gaile Moe, Danita Kelley, Jacqueline Berning and Carol Byrd-Bredbenner (2013). Wardlaw's Perspectives in Nutrition: A Functional Approach. McGraw-Hill, Inc., NY, USA.
- Swaminadhan M (1995). Principles of nutrition and dietics. Bappco.
- Tom Brody (1998). Nutritional Biochemistry (2nd ed), Academic press, USA.
- Garrow JS, James WPT and Ralph A (2000). Human nutrition and dietetics (10th ed) Churchill Livingstone.
- Andreas M Papas (1998). Antioxidant Status, Diet, Nutrition, and health (1st ed).
- CRC Branen, AL, Davidson PM A Salminen S (2001). Food Additives.2nd Ed. Marcel Dekker.
- Gerorge AB (1996). Encyclopaedia of food and colour Additives. Vol. III. CRC Press.
- Fatih Yildiz (2010). Advances in food biochemistry, CRC Press, Boca Raton, USA,
- Hui YH (2006). Food biochemistry and food processing, (Editor), Blackwell Publishing, Oxford, UK.
- Geoffrey Campbell-Platt (2009). Food Science and Technology. Wiley-Blackwell, UK.

WEB RESOURCES

- http://old.noise.ac.in/SecHmscicour/english/LESSON O3.pdf
- https://study.com/academy/lesson/energy-yielding-nutrients-carbohydratesfat-protein.html.
- https://www.nhsinform.scot/healthy-living/food-and-nutrition/eatingwell/vitamins-and-minerals

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THIRD YEAR	MAJOR BASED ELECTIVE - I	CREDIT : 4
SEMESTER -V	BIOTECHNOLOGY	25BCMB1B

COURSE OUTCOMES

The main objectives of this course are to

- Impart knowledge on gene manipulation and gene transfer technologies
- Make the students understand the procedures involved in plant tissue culture.
- Acquire knowledge on animal cell culture and stem cell technology.
- Improve the employability skills of students by providing knowledge in recent techniques such as PCR, blotting, ELISA etc.
- Understand the application of fermentation technology.

COURSE SPECIFIC OUTCOMES

- Acquire knowledge on rDNA technology, DNA manipulation, and use of restriction endonuclease.
- Get acquainted with the use of cloning and vectors in plant tissue culture.
- Understand the methods for production of proteins using recombinant DNA technology and their applications, basics of tissue culture, transgenes are, stem cell technology, risks, and safety aspects and patenting in biotechnology.
- Gain knowledge about the importance of gene and gene manipulation technologies.
- Know the concept fermentation technology and its applications.

UNIT - I **Hours: 12**

RECOMBINANT DNA TECHNOLOGY

Recombinant DNA technology - Principles of gene cloning: restriction endonucleases and other enzymes used in manipulating DNA molecules. Ligation of DNA molecules, DNA ligase, linkers and adapters, homopolymer tailing. End labelling and construction maps of PBR322, λ bacteriophage.

UNIT - II **Hours: 12**

PLANT TISSUE CULTURE

Plant tissue culture- basic requirements for culture, M S medium, callus culture, protoplast culture. Vectors – Ti plasmid (co-integration vector and binary vector), Viral vectors- TMV, CaMV and their applications. Transgenic plants – pest resistant, herbicide resistant and stress tolerant plants.

UNIT - III **Hours: 12**

ANIMAL TISSUE CULTURE

Animal cell lines and organ culture - culture methods and applications. Transgenic animals: transgenic mice- Production and its applications. Stem cell technology: definition, types, and applications.

UNIT - IV **Hours: 12**

MOLECULAR TECHNIQUES

PCR –Principle, types and its application in clinical diagnosis and forensic science. Southern blotting, Northern blotting and DNA finger printing Technique-principle and their applications.

UNIT - V **Hours: 12**

FERMENTATION TECHNOLOGY

Fermentation technology – Fermentors - general design, fermentation processes - Media used, downstream processing. Production and applications of ethanol, Streptomycin and Proteases. Production of edible vaccines.

REFERENCE BOOKS

1. Watson JD, Amy A Caudy, Richard M Myers and Jan Witkowski (2006). Recombinant DNA: Genes and Genomes - a Short Course (3rd ed), W.H. Freeman & Co
2. Satyanarayana U (2008). Biotechnology, Books & Allied (P) Ltd.
3. Cassida L (2007). Industrial Microbiology, New Age International.
4. Reed G (2004). Prescott and Dunn's Industrial Microbiology, CBS Publishers & Distributors.
5. David P Clark and Pazdernik NJ (2009). Biotechnology: applying the genetic Revolution-Elsevier.
6. Click BR and Pasternark JJ (2010). Molecular Biotechnology: Principles and Applications of Recombinant DNA. (4th ed) American Society for Microbiology.

WEB SOURCES

1. NPTEL Certification course - Gene Therapy by Sachin Kumar https://nptel.ac.in/courses/102/103/102103041.
2. Coursera Certification course –Vaccines.
3. https://futureoflife.org/background/benefits-risks-biotechnology.
4. https://www.sciencedirect.com/topics/neuroscience/genetic-engineering.
5. http://www.biologydiscussion.cm/biotechnology/techniquesbiotechnology/important-techniques-of-biotechnology-3-techniques/15683.
6. https://iopscience.iop.org/book/978-0-7503-1347-6/chapter/bk978-0-7503-1347-6ch1.
7. https://www.slideshare.net/zeal_eagle/fermentation-technology.
8. https://www.slideshare.net/zeal_eagle/fermentation-technology.
9. https://www.slideshare.net/Chepkitwai/blotting-techniques-6129300.

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THIRD YEAR	SKILL BASED ELECTIVE COURSE - I	CREDIT : 2
SEMESTER - V	BIOMEDICAL INSTRUMENTATION	25BCSB1A

COURSE OUTCOMES

The objectives of this course are to

- Provide insights about the blood pressure and its measurement.
- Elaborate the mechanism of instruments related to respiration.
- Highlight the importance of imaging techniques.
- Acquaint students about the basics of medical assisting devices.
- Familiarize about the life saving therapeutic equipments.

COURSE SPECIFIC OUTCOMES

- Illustrate the functions of instruments used for measuring blood pressure.
- Elaborate the devices required for monitoring of respiratory gases.
- Understand the operation of the imaging and sonographic instruments.
- Differentiate between the action of pacemakers, defibrillators and ventilators.

UNIT – I **Hours: 6**

Measurement of blood pressure – sphygmomanometer. Cardiac output – Cardiac rate – Heart sound – Stethoscope, ECG – EEG – EMG – ERG.

UNIT – II **Hours: 6**

Monitoring of inspired/expired anaesthetic gases, capnograph, inhalators, nebulizers, aspirators, infant respirator, Plethysmography.

UNIT – III **Hours: 6**

Medical imaging: X-ray machine - Radio graphic and fluoroscopic techniques – Computed tomography – MRI – PET, Ultrasonography – Endoscopy – Thermography.

UNIT – IV **Hours: 6**

Assisting equipments: Pacemakers – Defibrillators – Ventilators.

UNIT – V **Hours: 6**

Therapeutic equipments: Nerve and muscle stimulators –Diathermy – Heart – Lung machine – Audio meters – Dialyzers.

REFERENCE BOOKS

- Arumugam M. Bio-Medical Instrumentation, Anuradha Agencies.
- Geddes LA and Baker LE, Principles of Applied Bio-Medical Instrumentation, John
- Wiley and Sons J. Webster, Medical Instrumentation, John Wiley & Sons.
- Rajarao C and. Guha SK. Principles of Medical Electronics and Bio-medical Instrumentation, Universities (India) Ltd, Orient Longman Ltd.
- Leslie Cromwell, Fred J. Weibell and Erich A. Pfeiffer (2002). Bio-Medical Instrumentation and measurements, II Edition, Pearson Education,
- Khandpur RS. Handbook of Bio-Medical instrumentation, Tata McGraw Hill publishing Co Ltd.,

WEB RESOURCES

https://youtu.be/GkUCmb0cKwo?list=PLCZ9KmODEcu138IlVeHClJ4nskArYr1Dg

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THIRD YEAR	SKILL BASED ELECTIVE COURSE - I	CREDIT : 2
SEMESTER - V	BASICS OF FORENSIC SCIENCE	25BCSB1B

COURSE OUTCOMES

The main objectives of this course are to:

- Gain knowledge on the basic practices of forensic analysis.
- Perform investigation using fresh blood.
- Carry out the analysis using body fluids
- Investigate the presence of forms of drugs and poisons in body fluids.
- Execute the identification test on multiple samples.

COURSE SPECIFIC OUTCOMES

- Gain knowledge on basics of forensic science and method for collection and preservation of samples.
- Assess the paternity, maternity problems and DNA profiling.
- Identify the presence of alcohol, insecticides and pesticides in body fluids.
- Detail on the test performed to identify the presence of drugs and poisons in body fluids
- Identify species and sex from the available body fluids.

UNIT - I **Hours: 6**

Forensic Science: Definition, History and Development. Crime scene management and investigation; collection, preservation, packing and forwarding of physical and trace evidences for analysis.

UNIT II **Hours: 6**

Blood - grouping and typing of fresh blood samples including enzyme. Cases of disputed paternity and maternity problems, DNA profiling.

UNIT III **Hours: 6**

Analysis of body fluids- Analysis of illicit liquor including methyl and ethyl alcohol in body fluids and breathe. Chemical examination, physiology and pharmacology of Insecticides and pesticides.

UNIT IV **Hours: 6**

Psychotropic drugs -Sedatives, stimulants, opiates and drugs of abuse. Identification of poisons from viscera, tissues and body fluids.

UNIT V **Hours: 6**

Identification tests- Identification of hair, determination of species origin, sex, site and individual identification from hair. Classification and identification of fibres. Examination and identification of saliva, milk, urine and faecal matter

REFERENCE BOOKS

1. Norah Rudin and Keith Inman. An Introduction to Forensic DNA Analysis. USA, Second edition.
2. Saferstein and Richard E. Forensic Science Handbook, Volume 2 & 3.
4. Forensics by Embar-Seddon, Ayn and Pass. Allan D.
5. Forensic Medicine by Adelman, Howard C & Kobilinsky, Lawrence

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THIRD YEAR		CREDIT : 2
SEMESTER -V	SOFT SKILLS DEVELOPMENT	25SSUGC4

COURSE OUTCOMES

The main objectives of this course are to

- To develop the personality skills suited for corporate sector.
- To enable students to become competent and committed professionals to perform all IT
- enabled services effectively.
- To develop the soft skills and IT skills to serve in the Front Office Administration.
- To enable the students to perform in BPOs & Hospitality services.
- To enable the students to manage the Customer Care Services.
- To develop Computing skills to perform Data Entry and Office Automation Services.

COURSE SPECIFIC OUTCOMES

- To develop communicative competence among the Students.
- To enhance the learner's soft skills by giving adequate exposure in LSRW and sub skills.
- To enable learners to put the life skills into practice with confidence.

UNIT- I **Hours 6**

Know Thyself / Understanding Self: Introduction to Soft skills- Self-discovery-Developing positive attitude-Improving perceptions- Forming values.

UNIT- II **Hours 6**

Interpersonal Skills/ Understanding Others: Developing interpersonal relationship-Team building-group dynamics-Networking improved work relationship.

UNIT- III **Hours 6**

Communication Skills / Communication with Others: Art of listening –Art of reading –Art of speaking –Art of Writing-Art of writing emails e mail etiquette.

UNIT- IV **Hours 6**

Corporate Skills / Working with Others: Oral Presentation – Memos- Note taking - Note making and preparing Minutes- Reports, Proposals, Abstracts - Technical Writing.

UNIT- V **Hours 6**

Selling Self / Job Hunting: Writing resume/cv-interview Skills-Group discussion-Mock Interview-Mock GD – Goal setting - Career planning.

REFERENCE BOOKS

- Krishnasamy N, Manju Dhariwel and Lalitha Krishnasamy (2015). Mastering Communication Skills and Soft Skills – Bloomburg.
- Meena K and V Ayothi (2013). A Book on Development of Soft Skills (Soft Skills: A Road Map to Success), PR Publishers & Distributors,
- Meera Banerjee and Krishna Mohan: Developing Communication Skills, Trinity Publishers (Lakshmi Publications.
- Alex K. (2012) Soft Skills – Know Yourself & Know the World, S. Chand and Company LTD, Ram Nagar, New Delhi- 110 055.

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SEMESTER – VI

THIRD YEAR	CORE COURSE - VIII	CREDIT : 6
SEMESTER -VI	IMMUNOLOGY	25BCCC8

COURSE OUTCOMES

The main objectives of this course are to

- Introduce the structure and functions of lymphoid organs and cells of the immune system.
- Illustrate the structure and classification of antibodies and adaptive immune response.
- Impart knowledge on the types of immunity and uses of vaccines.

COURSE SPECIFIC OUTCOMES

- Provide an understanding of immune related diseases and transplantation.
- Study the Ag-Ab interaction and immunological techniques to identify antigens and antibodies.

UNIT – I **Hours 18**

Structure and function of primary lymphoid organs (thymus, bone marrow), secondary lymphoid organs (spleen, lymph node), Cells involved in immune system- Functions Phagocytosis –Inflammation.

UNIT – II **Hours 18**

Antigens - Nature, Immunogens, haptens, cross reactions - Immunoglobulin- types- structure and function. Cells involved in antibody formation, Clonal selection theory, Cooperation of T-cell with B-cell. Differentiation of T and B lymphocyte -Humoral and cell mediated immunity. Monoclonal antibody - Production and application in biology.

UNIT –III **Hours 18**

Immunity and its types - Innate, Acquired, active and passive- Natural and Artificial - Commonly used toxoid vaccines, killed vaccines, live attenuated vaccines, rDNA Vaccines, DNA and subunit vaccines.

UNIT - IV **Hours 18**

Hypersensitivity – Immediate (Type 1) and Delayed (Type IV), Auto- immune diseases with examples. Organ specific and systemic autoimmunity. SLE, RA. Transplantation – Types of Grafts, structure & functions of MHC, graft Vs host reaction, immunosuppressive Agents.

UNIT – V **Hours 18**

Antigen - antibody reactions, General features of Antigen Antibody reactions. Precipitation, Immuno diffusion, Radio immunodiffusion, Ouchterlony double diffusion, CIE, Rocket electrophoresis, Agglutination-Coomb's test Complement Fixation Test-Wasserman's reaction, RIA, ELISA.

REFERENCE BOOKS

- Kuby, J. (2018). Immunology (5th ed). W.H. Freeman - ISBN-10: 1319114709 / ISBN-13: 978-1319114701.
- Rao, C. V. (2017). Immunology (3rd ed). Chennai: Alpha Science Int. Ltd - ISBN-10: 1842652559/ ISBN 13:978-1842652558.
- Tizard (1995). An Introduction to Immunology. Harcourt Brace College Publications Kenneth Murphy M, Paul Travers and Mark Walport (2007), Janeway's Immunobiology, 7th edition, Garland Science.

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THIRD YEAR	CORE COURSE - IX	CREDIT : 5
SEMESTER -VI	CLINICAL BIOCHEMISTRY	25BCCC9

COURSE OUTCOMES

The main objectives of this course are to

- Comprehend the basic concepts and disorders of carbohydrate metabolism
- Explain the disorders of lipid metabolism.
- Elucidate the liver function test and kidney function test.

COURSE SPECIFIC OUTCOMES

- Designate the gastric function test.
- Familiarize the clinical enzymology.

UNIT – I **Hours 18**

Disorders of carbohydrate metabolism: Maintenance of blood glucose by hormone with special reference to insulin and glucagon. Abnormalities in glucose metabolism: Diabetes mellitus; types, causes, biochemical manifestations, diagnosis and treatment, glycated hemoglobin, hypoglycemia. Inborn errors of carbohydrate metabolism, glycosuria, Fructosuria, Pentosuria, Galactosemia and Glycogen storage diseases.

UNIT – II **Hours 18**

Disorders of amino acid metabolism- aminoaciduria, Phenylketonuria, Hartnup disease, alkaptonuria, albinism, cystinuria, cystinosis, homocystinuria and maple syrup urine Disease. Disorders of protein metabolism- Marasmus and kwashiorkor.

UNIT – III **Hours 18**

Disorders of Lipid Metabolism: Lipid Profile, Atherosclerosis, Fatty liver and hyper lipidemia. Hypercholesterolemia, Lipidosis and Xanthomatosis, Tay-Sach`s disease, Niemann Pick disease, lipotropic agents

UNIT – IV **Hours 18**

Liver Function Tests: Bilirubin metabolism and jaundice, Estimation of conjugated and total bilirubin in serum (Diazomethod). Detection of bilirubin and bile salts in urine (Fouchet's test and Hay's Sulphur test). Thymol turbidity test, prothrombin time, serum enzymes in liver disease serum transaminases (SGPT & SGOT) and lactate dehydrogenase (LDH).

Kidney Function Tests: Measurement of urine pH, volume, specific gravity, osmolality, sediments in urine, inulin, urea and creatinine clearance tests. Concentration and dilution tests. Phenol red test. Levels of plasma protein and its significance related to kidney function. Proteinuria.

UNIT – V **Hours 18**

Gastric Function test: Composition of gastric juice, collection of gastric contents, examination of gastric residuum, fractional test meal (FTM), stimulation test-alcohol and histamine stimulation, Tubeless gastric analysis.

REFERENCE BOOKS

- MN Chatterjee and Rana Shinde (2012). Text Book of Medical Biochemistry, Jaypee Brothers. Medical Publishers (P) LTD, New Delhi, 8th Edition,
- Ambika Shanmugam's Biochemistry for medical students, 8th edition, Published by Wolters Kluwer India Pvt. Ltd.

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THIRD YEAR	CORE PRACTICAL - VI	CREDIT : 4
SEMESTER -VI	CLINICAL BIOCHEMISTRY AND IMMUNOLOGY (P)	25BCCP6

COURSE OUTCOMES

The main objectives of this course are to

- Introduce the methods of sample collection (blood & urine) for analytical purpose.
- Impart practical knowledge on the assay of activity of various diagnostically important enzymes

COURSE SPECIFIC OUTCOMES

- Understand the estimation procedure for various important biomolecules.
- Help students learn the routine qualitative analysis of urine sample for diagnostic purpose.
- Train students on various hematological tests and its significance.

PRACTICALS HOURS: 60

1. Collection and preservation of urine samples.
2. Estimation of creatinine by Jaffe's method (serum & urine)
3. Estimation of urea by diacetyl monoxime method (serum & urine)
4. Estimation of cholesterol by Zak's method
5. Estimation of Glucose by Ortho Toluidine method
6. Estimation of Protein by Lowry's method
7. Qualitative analysis of normal constituents of urine (Urea, Creatinine, Phosphorus, Calcium)
8. Qualitative analysis of abnormal constituents of urine

a) Sugar (Glucose, fructose, pentose)
b) Protein
c) Amino acids (Tyrosine, Histidine, Tryptophan)
d) Ketone bodies
e) Bile pigments with clinical significance.

9. Blood smear identification of leucocytes by Giemsa stain.
10. Serum and plasma separation, Storage.
11. Precipitin reaction by double immunodiffusion and radial immunodiffusion (Ouchterlony and Mancini's methods).
12. Separation of mononuclear cells by Ficoll-Hypaque.

REFERENCE BOOKS

- Manickam SS (2018). Biochemical Methods (3rd ed.). New age International Pvt Ltd publishers - ISBN 10: 8122421407 / ISBN 13: 9788122421408.
- Plummer DT (n.d.). An Introduction to Practical Biochemistry. Tata McGraw Hill – ISBN: 97800708416.

- Alan H Gowenlock (1998). Varley's Practical Clinical Biochemistry, 6th edition, CBS Publishers, India.
- Godkar B (2020). Textbook of Medical Laboratory Technology Vol 1 & 2 Paperback, 3rd edition, Bhalani Publishers.
- Kanai L Mukerjee (1996). Medical Lab Technology, Vol I& II, 1st edition, Tata McGraw Hill, Pennsylvania.
- Ranjna Chawla (2014). Practical Clinical Biochemistry Methods and interpretations 58 (Paperback). 4th edition, Jaypee Brothers Medical Publishers, New York.
- Ranjna Chawla (2014). Practical Clinical Biochemistry Methods and interpretations 58 (Paperback). 4th edition, Jaypee Brothers Medical Publishers, New York.
- Singh SK (2005). Introductory Practical Biochemistry (2nd ed.). Alpha Science International, Ltd- ISBN 10: 8173193029 / ISBN 13: 9788173193026.
- Ashwood BA (2001). Tietz Fundamentals of Clinical chemistry. WB Saunders Company, Oxford Science Publications USA - ISBN 10: 0721686346 / ISBN 13: 978072168634.
- Talwar GP and Gupta SK. Practical and clinical immunology. 2nd edition.
- Asim Kumar Roy. Immunology Theory and Practical.

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THIRD YEAR	MAJOR BASED ELECTIVE COURSE - II	CREDIT : 4
SEMESTER -VI	ENDOCRINOLOGY	25BCMB2A

COURSE OUTCOMES

The main objectives of this course are to

- To enable students to learn endocrinology with special emphasis on the human endocrine system covering the anatomy, physiology and biochemistry of the system, biological phenomenon at cellular level.
- To provide detailed information on the release, effect and functioning of hormones.
- To acquire knowledge about the role of hormones as therapeutic agents.
- To acquaint students with experimental skills used in clinical and research Laboratories.

COURSE SPECIFIC OUTCOMES

- Comprehend the endocrine system and properties of hormones.
- Understand the importance of endocrine system and its role in maintenance of homeostasis.
- Gain in-depth knowledge of the molecular mechanism of hormone action and its regulation.
- Better appreciate the regulation of physiological process and its implication in diseases.
- Acquire information about human endocrine disorders.

UNIT - I **Hours: 15**

Introduction to the endocrine system and major glands (pituitary, pineal, adrenal, thyroid, parathyroid, testis, pancreas, ovaries, and GI tract), Classes of hormones, Modes of hormone secretion.

UNIT - II **Hours: 15**

Neuroendocrinology: General organization of nervous system and neuroendocrine organs; Neurons: Structure, types, distribution and characteristics; Introduction to Neuropeptides, Neurosteroids and neurohormones.

The hypothalamo-hypophyseal axis; Hypothalamo-vascular system; hypothalamic hormones: chemistry, physiology and its regulation. Hypothalamo-hypophyseal interactions with the gonads, adrenal and other endocrine glands. Neuroendocrine regulation of immune system; Stress hormones and immune response. Neuroendocrine disorders: genetic versus environmental causes (sleep apnea, precocious puberty).

UNIT - III **Hours: 15**

Molecular Endocrinology: Hormones as chemical messengers for control and regulation of physiological processes. Structure and biosynthesis of peptide, protein and steroid hormones; Storage, secretion and regulation of hormones; Mechanisms of hormone action: Receptor and non-receptor mediated signalling; Feedback mechanisms in signalling pathways.

UNIT - IV **Hours: 15**

Therapeutic use of hormones in health and disease (cancer, biological clock regulation, metabolic dysfunction, stress management, growth hormone disorders). Current developments in design and production of hormonal contraceptives. Recombinant protein hormones: production and application in regulation of fertility (Hormone replacement therapy, hypogonadism, PCOS/PCOD, xeno-estrogens and its effects on male fertility).

UNIT - V **Hours: 15**

Mineral Metabolism – Hormones and Disorders of Mineral Metabolism, Metabolic bone disease, Hypo & Hyper parathyroidism. Para endocrine syndromes, Hormonal manifestations of malignancy, Carcinoid tumours and carcinoid syndromes.

Endocrinology of HIV and AIDS. Gastro intestinal hormones and Gut hormones. Endocrine changes of pregnancy Endocrinology of aging, Immuno endocrinopathy syndrome.

REFERENCE BOOKS

- Williams Text book of Endocrinology
- Harrison's Textbook of Medicine
- Nelson's Textbook of pediatrics
- Degroot
- Pickup and Williams
- Joslin's Textbook of Diabetes
- David O. Norris, James Carr (2021) Vertebrate Endocrinology, V Edition, Elsevier.
- J. Larry Jameson, Leslie De Groot (2010). Endocrinology, VI Edition, Elsevier.
- Hadley, M.E. and Levine J.E. (2009). Endocrinology. VI Edition. Pearson Prentice Hall,
- Pearson Education Inc., New Jersey.
- Franklin F. Bolander (2004) Molecular Endocrinology. III Edition, Academic Press, USA.
- 196 Suggestive readings
- Handbook of Physiology published by American Physiological Society by Oxford University Press, Section 7: Multiple volumes set, 1998.
- Endocrinology: An Integrated Approach. BIOS Scientific Publishers (https://www.ncbi.nlm.nih.gov/books/NBK22/).
- Turner, D. (1977) General Endocrinology. VI Edition, Saunders.

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THIRD YEAR	MAJOR BASED ELECTIVE COURSE- II	CREDIT : 4
SEMESTER -VI	BIOINFORMATICS	25BCMB2B

COURSE OUTCOMES

The main objectives of this course are to

- Impart knowledge on bioinformatics and applications.
- Learn about biological databases.
- Understand the local and global sequence alignment.

COURSE SPECIFIC OUTCOMES

- Provide insights on BLAST and Microarray.
- Familiarize about structural genomics and visualization tools.

UNIT –I **Hours 15**

Bioinformatics – Bioinformatics and its applications. –Genome, Metabolome- Definition and its applications. Metabolome - Metabolome database - Human Metabolome database. Transcriptome - Definition and applications.

UNIT – II **Hours 15**

Biological Databases - definition, types and examples – Nucleotide sequence database (NCBI, EMBL, Genebank, DDBJ) Protein sequence database- SwissProt, TrEMBL, Structural Database-PDB.

UNIT – III **Hours 15**

Sequence Alignment - Local and Global alignment - Dot matrix analysis, PAM, BLOSUM. Dynamic Programming, Needleman - Wunch algorithm, Smith waterman algorithm.

UNIT – IV **Hours 15**

BLAST - features, types (BLASTP, BLASTN, BLASTX), PSI BLAST, result format. DNA Microarray-Procedure and applications.

UNIT – V **Hours 15**

Structural genomics - Whole genome sequencing (Shotgun approach), Comparative genomics-tools for genome comparison, VISTA servers and pre computed tools. Molecular visualization tools. RASMOL, Swiss PDB viewer.

REFERENCE BOOKS

- Basic of Bioinformatics by Rui Jiang Xuegong Zhang and Michael Q. Zhang Editors
- Bioinformatics for Beginners Genes, Genomes, Molecular Evolution, Databases and
- Analytical Tools by: Supratim Choudhuri (Author).
- Bioinformatics by Saras publication
- Introduction to Bioinformatics by Arthur Lesk
- S Balamurugan, Anand T. Krishnan, Dinesh Goyal, Balakumar Chandrasekaran Computation in BioInformatics Multidisciplinary Applications.
- Navneet Sharma. Chemoinformatics and Bioinformatics in the Pharmaceutical Sciences Himanshu Ojha, Pawan Raghav and Ramesh K. Goyal Pharmaceutics,

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THIRD YEAR	SKILL BASED ELECTIVE COURSE - I	CREDIT : 2
SEMESTER -VI	MEDICAL LAB TECHNOLOGY	25BCSB2A

COURSE OUTCOMES

- Impart knowledge on specimen collection and disposal of waste.
- Acquaint knowledge on collection, preservation and transfusion of blood.
- Quantify the biomolecules in biological sample.

COURSE SPECIFIC OUTCOMES

- Understand the significance of various tests and their interpretation in diseased conditions.
- Acquaint knowledge on enzymes, hormones and Immunoglobulins as markers for diagnosis.

UNIT-I **Hours 6**

Collection, transport, analysis of specimen – blood, routine urine, feces, sputum, semen, CSF Documentation of samples & results. Disposal of laboratory/ hospital Waste- Noninfectious waste, biomedical waste, infected sharp waste disposal, infected non sharp disposal – color coding as per guidelines.

UNIT-II **Hours 6**

Determination of Blood group and Rh factor -Basic blood banking procedures- cross matching, screening test. Blood transfusion and hazards.

UNIT-III **Hours 6**

Estimation of blood sugar – Enzymatic method, HbA1C, Qualitative and quantitative analysis of urine sample- NPN-urea, uric acid, creatinine. Mineral, vitamin and CSF analysis.

UNIT-IV **Hours 6**

Immuno diagnostics -Widal test, VDRL test, ASO, RA, CRP and Complement Fixation Test. RIA, ELISA, Skin test – Montaux and Lepramin test.

UNIT-V **Hours 6**

Assay of clinically important enzymes- Estimation of clinically important hormones – Insulin, Thyroid and Reproductive hormones and its clinical significance.

REFERENCE BOOKS

- Kanai L Mukherjee and Anuradha Chakravarthy (2022). Medical Laboratory Technology IV th edition, Vol I.
- Ramnik Sood (2006). Text Book of Medical Laboratory Technology, Jaypee Publishers,
- Tietz N (2018). Fundamentals of Clinical Chemistry and Molecular Diagnostics 8th edition, W.B. Saunders Company.

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THIRD YEAR	SKILLED BASED ELECTIVE COURSE - II	CREDIT : 2
SEMESTER -VI	MEDICAL CODING	25BCSB2B

COURSE OUTCOMES

The main objectives of this course are to

- Understand the basic concept of medical coding.
- Familiarize the student about medical terminology.

COURSE SPECIFIC OUTCOMES

- Understand about the classification of diseases based on WHO/AHA
- Understand about the CPT code used for diseases as per American Medical Association (AMA)

UNIT – I **Hours 6**

Introduction to Medical coding, coding theory, Healthcare Common Procedure Coding, First Aid and CPR.

UNIT – II **Hours 6**

Introduction to Medical Terminology, specialization I & II, Diagnostic coding, factors affecting diagnostic coding.

UNIT – III **Hours 6**

Documenting medical records- Importance of Documentation, Types of dictation formats.

UNIT – IV **Hours 6**

Introduction to Human Anatomy and Coding, ICD-10- CM classification system.

UNIT – V **Hours 6**

Introduction to CPT coding, types of CPT coding Medical Law and Ethics

REFERENCE BOOKS

- Understanding Medical Coding, A comprehensive guide Sandra L Johns on Robin Linker.
- Buck's Step – by – step Medical Coding Elsevier reference Terry Tropin M Shai, RHIA, CCS-P, AHIMAICD-10-CMcoding guidelines made easy2017.
- Besty J Shiland - Medical terminology and anatomy for ICD-10.

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THIRD YEAR		CREDIT : 1
SEMESTER -VI	GENDER STUDIES	25GSUGC5

COURSE OUTCOMES

- To explore existing gender biases in the society and to understand the need to work towards the inclusive society
- To inculcate sensitivity and build gender perspectives.

COURSE SPECIFIC OUTCOMES

- To use the course to bring attitudinal cum behavioural changes towards gender neutral ambience and promote the humanistic values.

UNIT- I **Hours 3**

Introduction to Gender Studies Concepts: Gender Spectrum-Sex – Gender distinction – Biological Determinism – Patriarchy – Feminism–Gender Socialization and Stereotyping-Gender Discrimination – Gender Division of labour and roles– Gender Sensitivity and awareness – Gender Equity – Equality – Gender Main streaming and Gender Analysis.

UNIT- II **Hours 3**

UGC Initiatives on Women's Studies: Definition of Women's Studies –Gender Studies –UGC Initiatives and guidelines on Women's Studies - Beijing Conference, UN Initiatives – Convention on Elimination of All forms of Discrimination Against Women (CEDAW)- Sustainable Development Goals on Gender Equality (SDG 5) and targets

UNIT- III **Hours 3**

Areas of Gender Discrimination: Gender Socialization- Sex Ratio– Health and Nutrition– – Literacy and Education - Employment- Governance – participation in decision making- politics- property rights and access to credit- gender based violence- Social institutions –Family, Caste, Class, religion, gender, State. Market – Media – Politics – Judiciary.

UNIT -IV **Hours 3**

Women Development and Gender Empowerment: Towards Equality Report of Status of Women in India 1974 – International Women's Decade – International Women's Year – National Policy for Empowerment of Women 2001

UNIT -V **Hours 3**

Women's Movements and Safeguarding Mechanism: In India National /State Commission for Women(NCW) – All Women Police Station – Family Court Legislations safeguarding women –Transgender Policy—Constitutional amendments for women's political participation.

REFERENCE BOOKS

- Bhasin Kamala (2004). Understanding Gender: Gender Basics, New Delhi: Women Unlimited.
- Bhasin Kamala (2004). Exploring Masculinity: Gender Basics, New Delhi: Women Unlimited.
- Bhasin Kamala (1993). What is Patriarchy? Gender Basics, New Delhi: Women Unlimited.
- Arya Sadhna (2000). Women, Gender Equality and the State, New Delhi: Deep &Deep Publication.
- Mishra OP (2001). Law Relating to Women & Child, Allahabad: Central Law Agency.
- Uma Chakravarti (2003). Gendering Caste Through a Feminist Lens, Sage Publication.

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SUBJECT CODE

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Nagore Post, Nagapattinam – 611 002, Tamilnadu.

PART - I				
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - Tamil	Tamil - I	25LCT1
II	I	Language Course - II (LC) - Tamil	Tamil - II	25LCT2
III	I	Language Course - III (LC) - Tamil	Tamil - III	25LCT3
IV	I	Language Course - IV (LC) - Tamil	Tamil - IV	25LCT4
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - French -	French - I	25LCF1
II	I	Language Course - I (LC) - French -	French - II	25LCF2
III	I	Language Course - I (LC) - French -	French - III	25LCF3
IV	I	Language Course - I (LC) - French -	French - IV	25LCF4
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - Arabic	Arabic - I	25LCA1
II	I	Language Course - I (LC) - Arabic	Arabic - II	25LCA2
III	I	Language Course - I (LC) - Arabic	Arabic - III	25LCA3
IV	I	Language Course - I (LC) - Arabic	Arabic - IV	25LCA4
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - Hindi	Hindi - I	25LCH1
II	I	Language Course - I (LC) - Hindi	Hindi - II	25LCH2
III	I	Language Course - I (LC) - Hindi	Hindi - III	25LCH3
IV	I	Language Course - I (LC) - Hindi	Hindi - IV	25LCH4
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - Sanskrit	Sanskrit - I	25LCS1
II	I	Language Course - I (LC) - Sanskrit	Sanskrit - II	25LCS2
III	I	Language Course - I (LC) - Sanskrit	Sanskrit - III	25LCS3
IV	I	Language Course - I (LC) - Sanskrit	Sanskrit - IV	25LCS4

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PART - II				
SEM	PART	COURSE	TITLE	SUBJECT CODE
I	II	English Language Course - I (ELC)	English - I	25ELC1
II	II	English Language Course - II (ELC)	English - II	25ELC2
III	II	English Language Course - III (ELC)	English - III	25ELC3
IV	II	English Language Course - IV (ELC)	English - IV	25ELC4
Common Papers				
SEM	PART	COURSE	TITLE	SUBJECT CODE
I	IV	Value Education	Value Education	25VEUGC1
II	IV	Environmental Studies	Environmental Studies	25ESUGC2
III	IV	Health and Wellness	Health and Well	25HWUGC3
V	IV	Soft Skills Development	Soft Skills Development	25SSUGC4
VI	V	Gender Studies	Gender Studies	25GSUGC5
VI	V	Extension Activities	Extension Activities	

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B.Sc Biochemistry

SEM	PART	COURSE	TITLE	SUBJECT CODE
I	III	Core Course - I (CC)	Biomolecules	25BCCC1
		Corse Practical – I (CP)	Biomolecules (P)	25BCCP1
		First Allied Course – I (AC)	Chemistry – I	25CHAC1
		First Allied Practical –I(AP)	Allied Chemistry –I (P)	25CHAP2
II	III	Core Course- II (CC)	Cell Biology	25BCCC2
		Core Practical – II (CP)	Cell Biology (P)	25BCCP2
		First Allied Course II(AC)	Chemistry – II	25CHAC2
		First Allied Practical – II (AP)	Allied Chemistry – II (P)	25CHAP2
III	III	Core Course-III (CC)	Biochemical Techniques	25BCCC3
		Core Practical – III (CP))	Biochemical Techniques (P)	25BCCP3
		Second Allied Course-I (AC)	Microbiology - I	25MBAC1
		Second Allied Practical –I (AP)	Allied Microbiology - I (P)	25MBAP1
IV	III	Core Course-IV (CC)	Human Physiology	25BCCC4
		Core Practical –IV (CC)	Human Physiology(P)	25BCCP4
		Second Allied Course – II(AC)	Microbiology – II	25MBAC2
		Second Allied Practical – II (AP)	Allied Microbiology - II (P)	25MBAP2
V	III	Core Course – V (CC)	Enzymes	25BCCC5
		Core Course – VI (CC)	Molecular Biology	25BCCC6
		Core Course – VII (CC)	Intermediary Metabolism	25BCCC7
		Core Practical – V (CP)	Enzymes & Molecular Biology (P)	25BCCP5
		Major Based Elective – I	(A)Nutritional Biochemistry	25BCMB1A
		Major Based Elective – I	(B)Biotechnology	25BCMB1B
		Internship	Internship	25BCIS
VI	III	Core Course –VIII (CC)	Immunology	25BCCC8
		Core Course – IX (CC)	Clinical Biochemistry	25BCCC9
		Core Practical– VI (CP)	Clinical Biochemistry & Immunology (P)	25BCCP6
		Major Based Elective Course –II	(A) Endocrinology	25BCMB2A
		Major Based Elective Course –II	(B) Bioinformatics	25BCMB2B
		Project	Project	25BCPW

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